16 a `OUTPUT T` should be `INPUT T`

`T < 30` should be `T < 20`

No flowline out of the `OUTPUT "Too Low"` box.

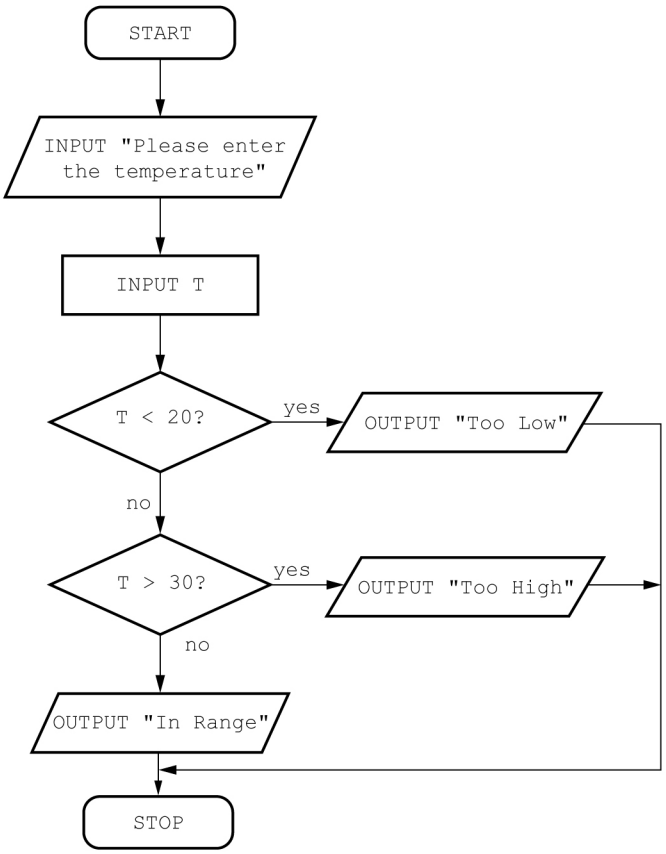
`T > 20` should be `T > 30`

Flowlines from the decision box are not labelled.

The arrow on the horizontal flowline is at the wrong end and points in the wrong direction.

No flowline out of the `OUTPUT "Too High"` box.

b



8 Programming

1 Three from:

- integer: a positive or negative whole number that can be used with mathematical operators
- real: a positive or negative number with a fractional part that can be used with mathematical operators
- Boolean: a variable or constant that can have only two values `TRUE` or `FALSE`
- char: a single alphanumeric character
- string: multiple alphanumeric characters

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2 a Python

```
Number1 = int(input("Please enter the first whole number "))
Number2 = int(input("Please enter the second whole number "))
Number3 = int(input("Please enter the third whole number "))
```

Visual Basic

```
Console.Write("Please enter first whole number ")
Number1 = Integer.Parse(Console.ReadLine())
Console.Write("Please enter second whole number ")
Number2 = Integer.Parse(Console.ReadLine())
Console.Write("Please enter third whole number ")
Number3 = Integer.Parse(Console.ReadLine())
```

Java

```
System.out.print("Please enter the first whole number ");
Number1 = myObj.nextInt();
System.out.print("Please enter the second whole number ");
Number2 = myObj.nextInt();
System.out.print("Please enter the third whole number ");
Number3 = myObj.nextInt();
```

b Python

```
Number1 = 10
Number2 = 20
Number3 = 30
```

Visual Basic

```
Number1 = 10
Number2 = 20
Number3 = 30
```

Java

```
Number1 = 10;
Number2 = 20;
Number3 = 30;
```

c Python

```
print("Integer values assigned and stored", Number1, Number2,
Number3)
```

Visual Basic

```
Console.WriteLine("Integer values assigned and stored " &
Number1 & " " & Number2 & " " & Number3)
```

Java

```
System.out.println("Integer values assigned and stored " +
Number1 + " " + Number2 + " " + Number3);
```


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d **Python**

```
Number1 = int(input("Please enter the first whole number "))
Number2 = int(input("Please enter the second whole number "))
Number3 = int(input("Please enter the third whole number "))

print("Integer values entered", Number1, Number2, Number3)

Number1 = 10
Number2 = 20
Number3 = 30

print("Integer values assigned and stored" , Number1,
Number2, Number3)
```

Visual Basic

```
Module Module1

    Sub Main()
        Dim Number1, Number2, Number3 As Integer
        Console.Write("Please enter first whole number ")
        Number1 = Integer.Parse(Console.ReadLine())
        Console.Write("Please enter second whole number ")
        Number2 = Integer.Parse(Console.ReadLine())
        Console.Write("Please enter third whole number ")
        Number3 = Integer.Parse(Console.ReadLine())

        Console.WriteLine("Integer values entered " & Number1
        & " " & Number2 & " " & Number3)

        Number1 = 10
        Number2 = 20
        Number3 = 30

        Console.WriteLine("Integer values assigned and stored
        " & Number1 & " " & Number2 & " " & Number3)

        Console.ReadKey()
    End Sub

End Module
```


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Java

```
import java.util.Scanner;
class Q2 Java
{
   public static void main(String args[])
   {
       Scanner myObj = new Scanner(System.in);

       int Number1, Number2, Number3;

       System.out.print("Please enter the first whole number ");
       Number1 = myObj.nextInt();
       System.out.print("Please enter the second whole number ");
       Number2 = myObj.nextInt();
       System.out.print("Please enter the third whole number ");
       Number3 = myObj.nextInt();

       System.out.println("Integer values entered  " +
       Number1 + " " + Number2 +" " + Number3);

       Number1 = 10;
       Number2 = 20;
       Number3 = 30;

       System.out.println("Integer values assigned and
       stored" + Number1 + " " + Number2 + " " + Number3);
    }
}
```

3 a **Python**

```
weight = 0
while weight < 0.5 or weight > 5.0:
    weight = float(input("Please enter the weight of your
    parcel "))
```

Visual Basic

```
weight = 0
While (weight < 0.5) Or (weight > 5.0)
    Console.Write("Please enter the weight of your parcel  ")
    weight = Decimal.Parse(Console.ReadLine())
End While
```


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Java

```
do {
  System.out.print("Please enter the weight of your parcel ");
  weight = myObj.nextDouble();
}
while (weight < 0.5 || weight > 5.0);
```

b **Python**

```
print("          Options")
print(" 1 Guaranteed next day delivery before noon")
print(" 2 Guaranteed next day delivery")
print(" 3 24-hour delivery")
print(" 4 48-hour delivery")
print(" 5 3-5 days delivery")
option = int(input("Please enter your chosen option "))

if option == 1:
    # option 1 code
elif option == 2:
    # option 2 code
elif option == 3:
    # option 3 code
elif option == 4:
    # option 4 code
elif option == 5:
    # option 5 code
else:
     print("Incorrect option chosen")
```

Visual Basic

```
Console.WriteLine("          Options")
Console.WriteLine(" 1 Guaranteed next day delivery before
noon")
Console.WriteLine(" 2 Guaranteed next day delivery")
Console.WriteLine(" 3 24-hour delivery")
Console.WriteLine(" 4 48-hour delivery")
Console.WriteLine(" 5 3-5 days delivery")

Console.Write ("Please enter your chosen option ")
opt = Integer.Parse(Console.ReadLine())
```


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```
Select Case opt
     Case 1
        'option 1 code
     Case 2
        'option 2 code
     Case 3
         'option 3 code
     Case 4
         'option 4 code
     Case 5
         'option 5 code
     Case Else
         Console.WriteLine("Incorrect option chosen")
End Select
```

Java

```
System.out.println("          Options");
System.out.println(" 1 Guaranteed next day delivery before
noon");
System.out.println(" 2 Guaranteed next day delivery");
System.out.println(" 3 24-hour delivery");
System.out.println(" 4 48-hour delivery");
System.out.println(" 5 3-5 days delivery");

System.out.print("Please enter your chosen option ");
option = myObj.nextInt();
switch (option) {
  case 1:
    //option 1 code ;
    break;
  case 2:
    //option 2 code;
    break;
  case 3:
    //option 3 code;
    break;
  case 4:
    //option 4 code;
    break;
```


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```
    case 5:
      //option 5 code;
      break;
    default:
      System.out.println("Incorrect option chosen");
  }
```

c **Python**

```
weight = 0
while weight < 0.5 or weight > 5.0:
    weight = float(input("Please enter the weight of your
    parcel "))

print("          Options")
print(" 1 Guaranteed next day delivery before noon")
print(" 2 Guaranteed next day delivery")
print(" 3 24-hour delivery")
print(" 4 48-hour delivery")
print(" 5 3-5 days delivery")
option = int(input("Please enter your chosen option "))

if option == 1:
    cost = weight * 10 + 1
elif option == 2:
    cost = weight * 10
elif option == 3:
    cost = 5
elif option == 4:
    cost = 4
elif option == 5:
    cost = 3
else:
    print ("Incorrect option chosen")
    cost = 0

print ("Your cost is ", cost)
```


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Visual Basic

```
Module Module1

    Sub Main()
        Dim weight, cost As Decimal
        Dim opt As Integer
        weight = 0
        While (weight < 0.5) Or (weight > 5.0)
           Console.Write("Please enter the weight of your
           parcel ")
           weight = Decimal.Parse(Console.ReadLine())
        End While

        Console.WriteLine("          Options")
        Console.WriteLine(" 1 Guaranteed next day delivery
        before noon")
        Console.WriteLine(" 2 Guaranteed next day delivery")
        Console.WriteLine(" 3 24-hour delivery")
        Console.WriteLine(" 4 48-hour delivery")
        Console.WriteLine(" 5 3-5 days delivery")

        Console.Write("Please enter your chosen option ")
        opt = Integer.Parse(Console.ReadLine())

        Select Case opt
            Case 1
                cost = weight * 10 + 1
            Case 2
                cost = weight * 10
            Case 3
                cost = 5
            Case 4
                cost = 4
            Case 5
                cost = 3
            Case Else
                Console.WriteLine("Incorrect option chosen")
                cost = 0
        End Select
```


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```
        Console.WriteLine("Your cost is " & cost)

        Console.ReadKey()

    End Sub

End Module
```

Java

```
import java.util.Scanner;
class Q3Java
{
   public static void main(String args[])
   {
       Scanner myObj = new Scanner(System.in);
       double weight, cost;
       int option;

       do {
       System.out.print("Please enter the weight of your
       parcel ");
       weight = myObj.nextDouble();
       }
       while (weight < 0.5 || weight > 5.0);

       System.out.println("          Options");
       System.out.println(" 1 Guaranteed next day delivery
       before noon");
       System.out.println(" 2 Guaranteed next day delivery");
       System.out.println(" 3 24-hour delivery");
       System.out.println(" 4 48-hour delivery");
       System.out.println(" 5 3-5 days delivery");

       System.out.print("Please enter your chosen option ");
       option = myObj.nextInt();
       switch (option) {
         case 1:
           cost = weight * 10 + 1;
           break;
```


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```
            case 2:
              cost = weight * 10;
              break;
            case 3:
              cost = 5;
              break;
            case 4:
              cost = 4;
              break;
            case 5:
              cost = 3;
              break;
            default:
              System.out.println("Incorrect option chosen");
              cost = 0;
              }
          System.out.println("Your cost is " + cost);
        }
    }
```

d

Test data	Expected output	Actual output
0.4	Error message	
5.1	Error message	
−1	Error message	
0.5 Option 1	Your cost is 6	
0.5 Option 2	Your cost is 5	
0.5 Option 3	Your cost is 5	
5 Option 1	Your cost is 51	
5 Option 2	Your cost is 50	
5 Option 4	Your cost is 4	
1 Option 5	Your cost is 3	
1 Option 7	Incorrect option chosen	

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4 a Counting: keeping a count of the number of times an action is performed:

```
Counter = Counter + 1
```

b Iteration: a section of programming code can be repeated under certain conditions.

For example, a pre-condition loop:

Python

```
while TotalWeight < 100:
        TotalWeight = TotalWeight + Weight
```

Visual Basic

```
While TotalWeight < 100
      TotalWeight = TotalWeight + Weight
End While
```

Java

```
while (TotalWeight < 100)
{
  TotalWeight = TotalWeight + Weight;
}
```

c Selection: allowing the selection of different paths through the steps of a program:

Python

```
if Age > 17:
    print("You are an adult")
```

Visual Basic

```
If Age > 17 Then
   Console.WriteLine("You are an adult")
End If
```

Java

```
if (Age > 17) {
  System.out.println ("You are an adult");
}
```

d Sequence – the order that steps in a program are executed.

For example: `TotalWeight` and `Weight` must be assigned values before the WHILE loop is executed.

Python

```
TotalWeight = 0
while TotalWeight < 100:
     Weight = float(input("Enter the next weight: "))
     TotalWeight = TotalWeight + Weight
```


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Visual Basic

```
Dim Weight, TotalWeight As Decimal
TotalWeight = 0

While TotalWeight < 100
    Console.WriteLine("Please enter the weight of your parcel ")
    Weight = Decimal.Parse(Console.ReadLine())
    TotalWeight = TotalWeight + Weight
End While
```

Java

```
import java.util.Scanner;
class Q4{
      public static void main(String args[])
      {
            double TotalWeight = 0;
            double Weight;
            Scanner myObj = new Scanner(System.in);

            while (TotalWeight < 100)
            {
                  System.out.print("Please enter the weight of
                  your parcel ");
                  Weight = myObj.nextDouble();
                  TotalWeight = TotalWeight + Weight;
            }
      }
}
```

e String handling – strings are used to store text. Every string contains a number of characters, from an empty string which has no characters stored to a maximum number specified by the programming language. The characters in a string can be labelled by position number. The first character in a string can be in position zero or position one, depending on the language.

For example, finding the number of characters in a string:

Python

```
len(MyString)
```

Visual Basic

```
MyString.Length()
```

Java

```
MyString.length();
```


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5

```
// Algorithm to input ten positive numbers and total them,
// output the total and then average
   DECLARE A, B, C, D : INTEGER
   DECLARE D : REAL
   A ← 0
   B ← 0

   REPEAT
       REPEAT
           OUTPUT "Please enter a positive number "
           INPUT C
       UNTIL C > 0
       A ← A + C
       B ← B + 1
   UNTIL B = 10
   D ← A / B
   OUTPUT A, D
```

6 a **Python**

```
# Python program to total 10 numbers and find the average
Total = 0
Counter = 0  # Total and Counter initialised to zero

while Counter < 10:
    Number = int(input("Please enter a positive number "))
    while Number <= 0:  # error trapping for negative numbers
        Number = int(input("Your number was zero or negative, please try
        again"))
    Total = Total + Number
    Counter = Counter + 1  # updating Total and Counter

Average = Total / Counter  # calculating the Average

print(" The total is ", Total, " the average is ", Average)
# outputting the results
```


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Visual Basic

```
'Visual Basic program to total 10 numbers and find the average

Module Module1

    Sub Main()
        Dim Total, Number, Counter As Integer
        Dim Average As Decimal ' variables declared

        Total = 0
        Counter = 0 ' Total and Counter initialised To zero

        Do
            Console.Write("Please enter a positive number ")
            Number = Int(Console.ReadLine())

            While (Number <= 0) 'error trapping for negative numbers
              Console.Write("Your number was zero or negative. Please try
              again ")
              Number = Int(Console.ReadLine())
            End While

            Total = Total + Number
            Counter = Counter + 1  'updating Total And Counter
        Loop Until Counter = 10

        Average = Total / Counter  ' calculating the Average

        Console.WriteLine(" The total is " & Total & " the average is " &
        Average) ' outputting the results

        Console.ReadLine()

    End Sub

End Module
```


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Java

```
// Java program to total 10 numbers and find the average
import java.util.Scanner;
class Q6Java
{
   public static void main(String args[])
   {
       Scanner myObj = new Scanner(System.in);

       int Counter, Number;
       double Average Total; //variables declared , Total must be double
       otherwise the division will be truncated

       Total = 0;
       Counter = 0; //Total and Counter initialised To zero

       do {

            System.out.print("Please enter a positive number ");
            Number = myObj.nextInt();

            while (Number <= 0){

                System.out.print("Your number was zero or negative. Please
                try again ");
                Number = myObj.nextInt(); //error trapping for negative
                numbers
            }

            Total = Total + Number;
            Counter = Counter + 1; //updating Total And Counter
        }
        while (Counter < 10);

        Average = Total / Counter ; // calculating the Average

        System.out.println(" The total is " + Total + " the average is " +
        Average);//outputting the results
    }
}
```


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b

Total	Number	Counter	Average	OUTPUT
0		0		
4	4	1		Please enter a positive number
7	3	2		Please enter a positive number
14	7	3		Please enter a positive number
20	6	4		Please enter a positive number
	0			Your number was negative or zero, please try again
	-3			Your number was negative or zero, please try again
29	9	5		Please enter a positive number
30	1	6		Please enter a positive number
38	8	7		Please enter a positive number
43	5	8		Please enter a positive number
45	2	9		Please enter a positive number
50	5	10		Please enter a positive number
			5.0	The total is 50 the average is 5.0

7 a **Python**

```
Student = ["Alice", "Jan", "Lim", "Ian", "Min", "Sue", "Tim",
"Dee"]
for Counter in range(8):
    print(Student[Counter])
print()

Counter = 0
while Counter < 8:
    print(Student[Counter])
    Counter = Counter + 1
print()
```

Visual Basic

```
For Counter = 0 To 7
    Console.WriteLine(Student(Counter))
Next
Console.WriteLine()
```


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```
Counter = 0
Do
    Console.WriteLine(Student(Counter))
    Counter = Counter + 1
Loop Until Counter = 8

Console.WriteLine()
```

Java

```
for (Counter = 0; Counter <= 7; Counter++) {
     System.out.println(Student[Counter]);
}
System.out.println()

Counter = 0;
while (Counter <= 7){
      System.out.println(Student[Counter]);
      Counter++;
}

System.out.println()
```

b **Python**

```
NumberOfStudents = len(Student)

Counter = 0
while Counter < NumberOfStudents:
    print(Student[Counter])
    Counter = Counter + 1
print("Number of students", NumberOfStudents)
```

Visual Basic

```
NumberOfStudents = Student.Length
Counter = 0
Do
    Console.WriteLine(Student(Counter))
    Counter = Counter + 1
Loop Until Counter = NumberOfStudents

Console.WriteLine("Number of students " & NumberOfStudents)
```

Java

```
NumberOfStudents = Student.length;
Counter = 0;
while (Counter < NumberOfStudents){
    System.out.println("Number of students " + NumberOfStudents);
    Counter++;
}
```

8 **a** Find the length: `LENGTH(MyString)`

Convert to upper case: `UCASE(MyString)`

Convert to lower case: `LCASE(MyString)`

Find the first character: `SUBSTRING(MyString, 1, 1)`

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b **Python**

```
MyString = "Test String"
Length = len(MyString)
FirstChar = MyString[0:1]
UpperCase = MyString.upper()
LowerCase = MyString.lower()

print("Length of string ", Length)
print("First character of string ", FirstChar)
print("String in upper case ", UpperCase)
print("String in lower case ", LowerCase)
```

Visual Basic

```
Module Module1

    Sub Main()

        Dim MyString, FirstChar, UpperCase, LowerCase As String
        Dim Length As Integer
        MyString = "Test String"
        Length = MyString.Length()
        FirstChar = MyString.Substring(0, 1)
        UpperCase = UCase(MyString)
        LowerCase = LCase(MyString)

        Console.WriteLine("Length of string " & Length)
        Console.WriteLine("First character of string " & FirstChar)
        Console.WriteLine("String in upper case " & UpperCase)
        Console.WriteLine("String in lower case " & LowerCase)

        Console.ReadKey()
    End Sub

End Module
```

Java

```
import java.util.*;
class Q8Java
{
   public static void main(String args[])
    {
       Scanner myObj = new Scanner(System.in);


       String MyString, FirstChar, UpperCase, LowerCase;
       int Length;
       MyString = "Test String";
       Length = MyString.length();
       FirstChar = MyString.substring(0, 1);
       UpperCase = MyString.toUpperCase();
       LowerCase = MyString.toLowerCase();

       System.out.println("Length of string " + Length);
       System.out.println("First character of string " +
       FirstChar);
       System.out.println("String in upper case " + UpperCase);
       System.out.println("String in lower case " + LowerCase);

    }
}
```


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9 a **Python**

```
A = ((B + C - D * E) ** F) / G
```

Visual Basic

```
A = ((B + C - D * E) ^ F) / G
```

Java

```
A = Math.pow((B + C - D * E), F) / G;
```

b **Python**

```
if (A == B) or (C != D):
    A = 0
elif (A > B) and (C == D):
    B = 0
```

Visual Basic

```
If (A = B) Or (C <> D) Then
    A = 0
Else
    If (A > B) And (C = D) Then
        B = 0
    End If
End If
```

Java

```
if (A == B) or (C != D) {
    A = 0;
}
else {
    if ((A > B) and (C = D)) {
        B = 0;
    }
}
```

c

```
FOR Times = 1 TO 20
    FOR Count = 1 TO 10
        OUTPUT Count
    NEXT Count
NEXT Times
```

d **Python**

```
for Times in range(20):
    for Count in range(10):
            print(Count + 1)
```

Visual Basic

```
Module Module1

    Sub Main()
        Dim Count, Times As Integer
        For Times = 1 To 20
            For Count = 1 To 10
                Console.Write(Count)
            Next
            Console.WriteLine()
        Next

        Console.ReadKey()
    End Sub
End Module
```


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Java

```
import java.util.*;
class Q9Java
{
   public static void main(String args[])
    {

       int Count, Times;
       for (Times = 1; Times <= 20; Times++) {
           for (Count = 1; Count <= 10; Count++) {
           System.out.print(Count);
                }
           System.out.println();
       }

    }
}
```

10 Tasks that are repeated many times in an algorithm make use of **procedures** and **functions**. Use of these can reduce the size of a **program**.

Procedures and functions are **defined** once and **called** many times. They can be written with and without **parameters**. They are the variables that store the values passed to a procedure or function.

Functions always **return** a **value** and the value can be used on the right-hand side of an assignment statement.

A variable that can be used in any part of a program is called a **global** variable. A variable that is declared within a procedure or function is called a **local** variable.

11 a Library routines are fully tested and ready to be incorporated into a program …

… they perform many types of task that need to be used regularly.

b 1 MOD

2 DIV

3 ROUND

4 RANDOM

c **Python**

```
import random

Number1 = int(random.random() * 10 + 10)
Number2 = int(random.random() * 10 + 10)
print(Number1, "is the first random integer between 10 and 20")
print(Number2, "is the second random integer between 10 and 20")

AnswerMod = Number1 % Number2
AnswerDiv = Number1 // Number2

print(Number1, "MOD", Number2, "is", AnswerMod)
print(Number1, "DIV", Number2, "is", AnswerDiv)
```


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Visual Basic

```
Module Module1

    Sub Main()
        Dim Number1, Number2 As Integer
        Dim AnswerMod, AnswerDiv As Integer
        Randomize()
        Number1 = CInt(Rnd() * 10 + 10)
        Number2 = CInt(Rnd() * 10 + 10)
        Console.WriteLine(Number1 & " is the first random integer
        between 10 and 20")
        Console.WriteLine(Number2 & " is the second random integer
        between 10 and 20")

        AnswerMod = Number1 Mod Number2
        AnswerDiv = Number1 \ Number2

        Console.WriteLine(Number1 & " MOD " & Number2 & " is " &
        AnswerMod)
        Console.WriteLine(Number1 & " DIV " & Number2 & " is " &
        AnswerDiv)


        Console.ReadKey()

    End Sub

End Module
```

Java

```
import java.util.Scanner;
import java.lang.Math;
import java.util.Random;

class Q11Java
{
   public static void main(String args[])
    {
       Scanner myObj = new Scanner(System.in);
       int Number1, Number2;
       Random rand = new Random();
       Number1 = rand.nextInt(10) + 10;
       Number2 = rand.nextInt(10) + 10;

       System.out.println(Number1 + " is the first random integer
       between 10 and 20");
       System.out.println(Number2 + " is the second random
       integer between 10 and 20");
       int AnswerMod = Number1 % Number2;
       int AnswerDiv = Number1 / Number2;

       System.out.println(Number1 + " MOD " + Number2 + " is " +
       AnswerMod);
       System.out.println(Number1 + " DIV " + Number2 + " is " +
       AnswerDiv);

    }
}
```


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12 a array: a data structure containing several elements of the same data type; these elements can be accessed using the same identifier name

array dimension: dimension determined by how many index numbers are needed to locate a piece of data, for instance a one-dimensional array only needs one index number, a two dimensional array needs two index numbers, and so on

array index: identifies the position of an element in an array

b

```
DECLARE OXO : ARRAY[3, 3] OF CHAR

DECLARE PROCEDURE ShowGame()
    DECLARE RowIndex, ColumnIndex : INTEGER
    FOR RowIndex ← 1 TO 3
        PRINT OXO[RowIindex,1], OXO[RowIindex,2], OXO[RowIindex,3]
    NEXT
ENDPROCEDURE
```

c

Python

```
OXO =[["O"," "," "],[" ", "X", " "],[" ", "X", "X"]]
def PrintStateOfGame():
    print(OXO[0])
    print(OXO[1])
    print(OXO[2])

PrintStateOfGame()
```

Visual Basic

```
Module Module1
    Public OXO = New String(2, 2) {{"O", " ", " "}, {" ", "X", " "},
    {" ", "X", "O"}} 'OXO must be declared as a public/global variable

    Sub Main()
        PrintStateOfGame()
        Console.ReadKey()
    End Sub
    Sub PrintStateOfGame()
        Dim RowCounter, ColumnCounter As Integer
        For RowCounter = 0 To 2
            For ColumnCounter = 0 To 2
                Console.Write(OXO(RowCounter, ColumnCounter))
            Next
            Console.WriteLine()
        Next
    End Sub
End Module
```

Java

```
import java.util.*;

class Q12Java

{
    public static String[][] OXO = {{"O", " ", " "}, {" ", "X", " "},
    {" ", "X", "O"}};

    static void PrintStateOfGame()
    {
       int RowCounter, ColumnCounter;
       for (RowCounter = 0; RowCounter <= 2; RowCounter++){
           for (ColumnCounter = 0; ColumnCounter <= 2; ColumnCounter++){
               System.out.print(OXO[RowCounter][ColumnCounter]);
            }
       System.out.println();
    }
```


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```
}

    public static void main(String args[])
    {
              PrintStateOfGame();
    }
```

13 a Data that will be required again will be lost if it is only in RAM when the computer is switched off …

… if data is stored in a file, it can be accessed by the same or another program at a later date.

b i

```
DECLARE Text : STRING
DECLARE MyFile : STRING


MyFile ← "MyFile.txt"
Text ← "Test"
OPEN MyFile FOR WRITE
    WRITEFILE, Text
CLOSEFILE(MyFile)
```

ii

```
OPEN MyFile FOR READ
    READFILE, Text
    OUTPUT Text
CLOSEFILE(MyFile)
```

c

Python

```
MyFile = open("MyFile.txt","w")
Text = "Test"
MyFile.write(Text)

print("The file contains this line of text")
MyFile = open("MyFile.txt","r")
Text = MyFile.read()
print(Text)
```

Visual Basic

```
Imports System.IO
Module Module1
    Sub Main()
        Dim Text As String
        Dim objMyFileWrite As StreamWriter
        Dim objMyFileRead As StreamReader
        objMyFileWrite = New StreamWriter("MyFile.txt")
        Console.Write("Please enter a line of text  ")
        Text = "Test"
        objMyFileWrite.WriteLine(Text)
        objMyFileWrite.Close()

        objMyFileRead = New StreamReader("MyFile.txt")
        Text = objMyFileRead.ReadLine()
        Console.Write("The file contains this line of text ")
        Console.WriteLine(Text)
        objMyFileRead.Close()
    End Sub
End Module
```


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Java

```
import java.io.BufferedReader;
import java.io.PrintWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

class TextFile {

    public static void main(String[] args) {
        String Text;
        try {
            FileWriter myFileWriter = new FileWriter("MyFile.txt", false);
            PrintWriter myPrintWriter = new PrintWriter(myFileWriter);

            Text = "Test";
            myPrintWriter.printf("%s" + "%n", Text);
            myPrintWriter.close();
      } catch (IOException e) {
          e.printStackTrace();
      }

      try {
        FileReader myFileReader  = new FileReader("MyFile.txt");
        BufferedReader myBufferReader  = new BufferedReader(myFileReader);

        Text = myBufferReader.readLine();
        System.out.print("The file contains this line of text ")
        System.out.println(Text);

        myFileReader.close();

      } catch (IOException e) {
          e.printStackTrace();
      }
   }

}
```

14 a–e:

Python

```
#### Part b ###
def NewPassword():
    MyPassword = input("Please enter your new password ")
    Valid = True
    Length = len(MyPassword)
    if Length > 20:
        print("Too long")
        Valid = False
    if Length < 10:
        print("Too short")
        Valid = False
```


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```
Counter = 0
 NoSpace = True
 while (Counter < Length and NoSpace):
     if MyPassword[Counter:Counter + 1] == " ":
         print("Spaces not allowed")
         Valid = False
         NoSpace = False
     Counter = Counter + 1

Counter = 0
 UpperCase = False
 while (Counter < Length and not UpperCase):
     if "A" <= MyPassword[Counter:Counter + 1] <= "Z":
         UpperCase = True
     Counter = Counter + 1
 if (not UpperCase):
     print("No uppercase letter")
     Valid = False

 Counter = 0
 Digit = False
 while (Counter < Length and not Digit):
     if "0" <= MyPassword[Counter:Counter + 1] <= "9":
         Digit = True
     Counter = Counter + 1
 if (not Digit):
     print("No digit")
     Valid = False

if Valid:
     print("Password accepted")
     MyFile = open("MyPassword.txt","w")    #### Part c ###
     MyFile.write(MyPassword)               #### Part c ###
 else:
     print("Password rejected")
```


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```
#### Part d ###

def CheckPassword():
    MyPassword = input("Please enter your password ")
    MyFile = open("MyPassword.txt","r")
    StoredPassword = MyFile.read()
    if MyPassword == StoredPassword:
        print("Password matches")
    else:
        print("Password does not match")


#### Part e ###

def ChangePassword():
    MyPassword = input("Please enter your password ")
    MyFile = open("MyPassword.txt","r")
    StoredPassword = MyFile.read()

    if MyPassword == StoredPassword:
        NewPassword()
    else:
        print("Password incorrect cannot change")

#### Part a ###

Quit = False
while not Quit:
    print ("    Password Management")
    print (" 1.  Enter a new password")
    print (" 2.  Check your password")
    print (" 3.  Change your password")
    print (" 4.  Quit")

    option = int(input("Please enter your chosen option "))
```


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```
    if option == 1:
        NewPassword()
    elif option == 2:
        CheckPassword()
    elif option == 3:
        ChangePassword()
    elif option == 4:
        Quit = True
    else:
        print ("Incorrect option chosen")
```

Visual Basic

```
Imports System.IO
Module Module1

'*** Part a ***
    Sub Main()
        Dim Opt As Integer
        Dim Quit As Boolean = False
        Dim MyPassword As String

        While Not Quit
            Console.WriteLine("    Password Management")
            Console.WriteLine(" 1. Enter a new password")
            Console.WriteLine(" 2. Check your password")
            Console.WriteLine(" 3. Change your password")
            Console.WriteLine(" 4. Quit")

            Console.Write("Please enter your chosen option ")
            Opt = Integer.Parse(Console.ReadLine())
```


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```
            Select Case Opt
                Case 1
                    Console.Write("Please enter your password ")
                    MyPassword = Console.ReadLine()
                    NewPassword(MyPassword)
                Case 2
                    Console.Write("Please enter your password ")
                    MyPassword = Console.ReadLine()
                    CheckPassword(MyPassword)
                Case 3
                    Console.Write("Please enter your password ")
                    MyPassword = Console.ReadLine()
                    ChangePassword(MyPassword)
                Case 4
                    Quit = True
                Case Else
                    Console.WriteLine("Incorrect option chosen")
            End Select

            Console.ReadKey()

        End While

    End Sub
'******
```


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```
'*** Part b ***
    Sub NewPassword(MyString As String)
        Dim Length, Counter As Integer
        Dim Valid, NoSpace, UpperCase, Digit As Boolean
        Dim objMyFileWrite As StreamWriter

        Valid = True
        Length = Len(MyString)
        If Length > 20 Then
            Console.WriteLine("Too long")
            Valid = False
        End If
        If Length < 10 Then
            Console.WriteLine("Too short")
            Valid = False
        End If

        Counter = 0
        NoSpace = True
        While Counter < Length And NoSpace
            If MyString.Substring(Counter, 1) = " " Then
                Console.WriteLine("Spaces not allowed ")
                Valid = False
                NoSpace = False
            End If
            Counter = Counter + 1
        End While
```


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```
Counter = 0
UpperCase = False
While Counter < Length And Not UpperCase
    If MyString.Substring(Counter, 1) >= "A" And
    MyString.Substring(Counter, 1) <= "Z" Then
        UpperCase = True
    End If
    Counter = Counter + 1
End While
If Not UpperCase Then
    Console.WriteLine("No uppercase letter")
    Valid = False
End If

Counter = 0
Digit = False
While Counter < Length And Not Digit
    If MyString.Substring(Counter, 1) >= "0" And
    MyString.Substring(Counter, 1) <= "9" Then
        Digit = True
    End If
    Counter = Counter + 1
End While
If Not Digit Then
    Console.WriteLine("No digit")
    Valid = False
End If
If Valid Then
    Console.WriteLine("Password accepted. Press any
    key to continue.")
    objMyFileWrite = New
    StreamWriter("MyPassword.txt")'*** Part c ***
    objMyFileWrite.WriteLine(MyString)'*** Part c ***
    objMyFileWrite.Close()'*** Part c ***
Else
```


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```
            Console.WriteLine("Password rejected. Press any
            key to continue.")
        End If

    End Sub
'******

'**** Part d ***
    Sub CheckPassword(MyString As String)
        Dim objMyFileRead As StreamReader
        Dim StoredPassword As String
        objMyFileRead = New StreamReader("MyPassword.txt")
        StoredPassword = objMyFileRead.ReadLine
        objMyFileRead.Close()

        If MyString = StoredPassword Then
            Console.Write("Password matches. Press any key to
            continue.")
        Else
            Console.WriteLine("Password does not match. Press
            any key to continue.")
        End If

    End Sub
'******

'*** Part e ***
    Sub ChangePassword(MyString As String)
        Dim NewWord As String
        Dim objMyFileRead As StreamReader
        Dim StoredPassword As String
        objMyFileRead = New StreamReader("MyPassword.txt")
        StoredPassword = objMyFileRead.ReadLine()
        objMyFileRead.Close()
```


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```
        If MyString = StoredPassword Then
            Console.Write("Please enter your new password ")
            NewWord = Console.ReadLine()
            NewPassword(NewWord)
        Else
            Console.WriteLine("Password incorrect cannot
            change. Press any key to continue.")
        End If

    End Sub
'******

End Module
```

Java

```
import java.util.*;
import java.io.BufferedReader;
import java.io.PrintWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

class Chapter8Q14 {
 // *** Part b ***
    static void NewPassword(String MyString) {
        int Length;
        Length = MyString.length();
        boolean Valid = true;

        if (Length > 20) {
            System.out.println("Too long");
            Valid = false;
        }
```


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```
 if (Length < 10) {
     System.out.println("Too short");
     Valid = false;
 }

int Counter = 0;
boolean NoSpace = true;

while ((Counter < Length) & NoSpace) {
  if (MyString.substring(Counter,Counter +
  1).compareTo( " ") == 0) {
      System.out.println("Spaces not allowed");
      Valid = false;
      NoSpace = false;
  }
  Counter ++;

  }

Counter = 0;
boolean UpperCase = false;
while ((Counter < Length) & !UpperCase) {
   if (MyString.charAt(Counter) >= 'A' &&
   MyString.charAt(Counter) <= 'Z') {
       UpperCase = true;
   }
   Counter ++;
}
if(!UpperCase) {
    System.out.println("No uppercase letter");
    Valid = false;
}
```


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```
        Counter = 0;
        boolean Digit = false;
        while ((Counter < Length) & !Digit) {

           if (MyString.charAt(Counter) >= '0' &&
           MyString.charAt(Counter) <= '9') {
           Digit = true;
           }
           Counter ++;
        }
        if(!Digit) {
            System.out.println("No digit");
            Valid = false;
        }



        if (Valid) {
            System.out.println("Password accepted");
            try {
                 FileWriter myFileWriter = new
                 FileWriter("MyPassword.txt", false); //***
                 Part c ***
                 PrintWriter myPrintWriter = new
                 PrintWriter(myFileWriter); //*** Part c ***
                 myPrintWriter.printf("%s" + "%n", MyString);
                 //*** Part c ***
                 myPrintWriter.close(); //*** Part c ***
            } catch (IOException e) { //*** Part c ***
                 e.printStackTrace(); //*** Part c ***
            }
}
   else {
        System.out.println("Password rejected");
    }
}
```


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```
// ******

// *** Part d ***
   static void CheckPassword(String MyString) {
      String StoredPassword = "";

      try {
            FileReader myFileReader  = new
            FileReader("MyPassword.txt");
            BufferedReader myBufferReader  = new
            BufferedReader(myFileReader);
            StoredPassword = myBufferReader.readLine();
            myFileReader.close();
      } catch (IOException e) {
            e.printStackTrace();
      }

      if (MyString.compareTo(StoredPassword) == 0) {
            System.out.println("Password matches");
      }
      else {
            System.out.println("Password does not match");
      }

   }
//******
```


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```
// *** Part e ***
   static void ChangePassword(String MyString) {
      Scanner myObj = new Scanner(System.in);
      String StoredPassword = "";
      String NewWord = "";

      try {
            FileReader myFileReader  = new
            FileReader("MyPassword.txt");
            BufferedReader myBufferReader  = new
            BufferedReader(myFileReader);
            StoredPassword = myBufferReader.readLine();
            myFileReader.close();
      } catch (IOException e) {
            e.printStackTrace();
      }

      if (MyString.compareTo(StoredPassword) == 0) {
            System.out.print("Please enter your new password ");
            NewWord = myObj.nextLine();
            NewPassword(NewWord);
      }
      else {
            System.out.println("Password incorrect cannot
            change");
      }
   }
//******
```


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```
// *** Part a ***
   public static void main(String args[]) {
       Scanner myObj = new Scanner(System.in);
       boolean Quit = false;
       while (!Quit) {
            System.out.println("     Password Management");
            System.out.println(" 1   Enter a new password");
            System.out.println(" 2   Check your password");
            System.out.println(" 3   Change your password");
            System.out.println(" 4   Quit");

            System.out.print("Please enter your chosen option ");
            int option = myObj.nextInt();

            String MyPassword = myObj.nextLine();



           switch (option) {
               case 1:
                    System.out.print("Please enter your
                    password ");
                    MyPassword = myObj.nextLine();
                    NewPassword(MyPassword);
                    break;
               case 2:
                    System.out.print("Please enter your
                    password ");
                    MyPassword = myObj.nextLine();
                    CheckPassword(MyPassword);
                    break;
               case 3:
                    System.out.print("Please enter your
                    password ");
                    MyPassword = myObj.nextLine();
                    ChangePassword(MyPassword);
                    break;
               case 4:
                    Quit = true;
                    break;
```


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```
                    default:
                        System.out.println("Incorrect option
                        chosen");
                }
            }
        //******
        }
    }
```

b

Test data	Expected output	Actual output
Password	Too short No digit Password rejected	
Password99	Password accepted	
password99	No uppercase letter Password rejected	
Password99!	Password accepted	
ExtraLargePassword9999999	Too long Password rejected	
2 1	Too short Spaces not allowed No uppercase letter Password rejected	

15 Important points to note when writing your program:

- There is not one 'correct' solution to this problem. Any solution is acceptable but your program must:
  - perform all the tasks set out in the question
  - use a range of appropriate programming techniques that cover all the tasks set out in the question. These tasks must be executed by your program in a logical order
  - use appropriate data structures, which have been given meaningful names, to store all the data mentioned in the question. If any data structures are named in the question, then those names must be used in your program
  - be fully commented to ensure that it can be understood
- The array name must be `Game[]`
- All other data structures must have meaningful names.

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